Midterm Project- R Code and Outputs

Load in neccessary packages

```
library(tidymodels)
library(caret)
library(corrplot)
```

Part I

Using this dataset, please help Researcher A build a prediction model for PFS, aiming to understand how demographic, tumor, and treatment characteristics influence progression-free survival in breast cancer patients.

To answer this question, your report should at least include the following:

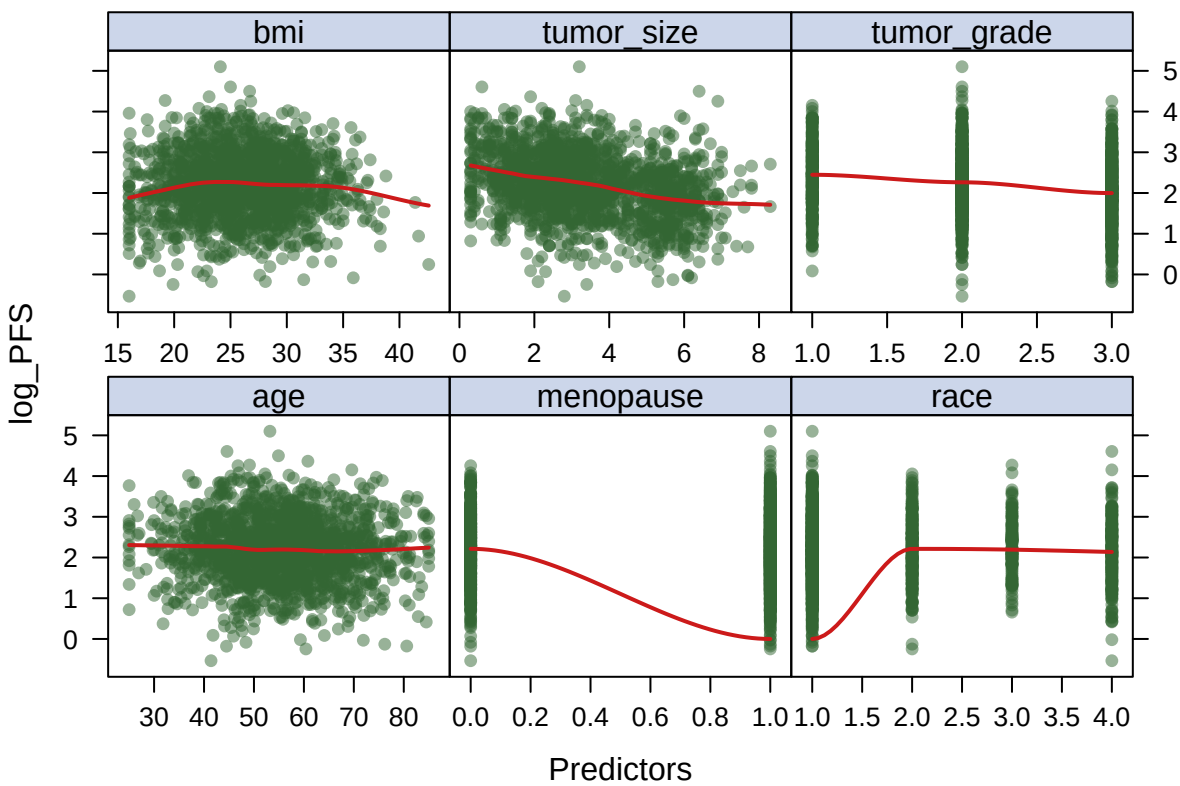
Model Training: Provide a detailed description of the model training procedure and how you obtained the model. Your description should include sufficient detail so that another statistician can reproduce the same model. Points will be deducted if the details provided are insufficient to reproduce your results.

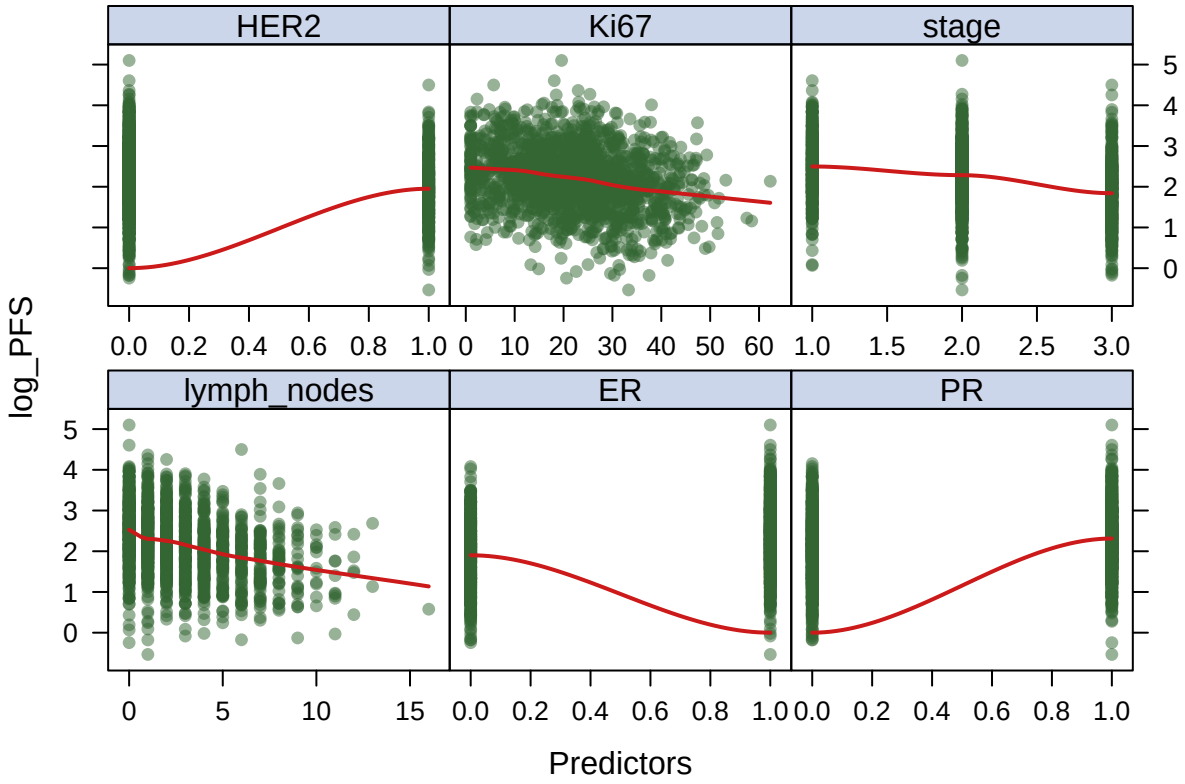
Import and partition data using 'tidymodels' package

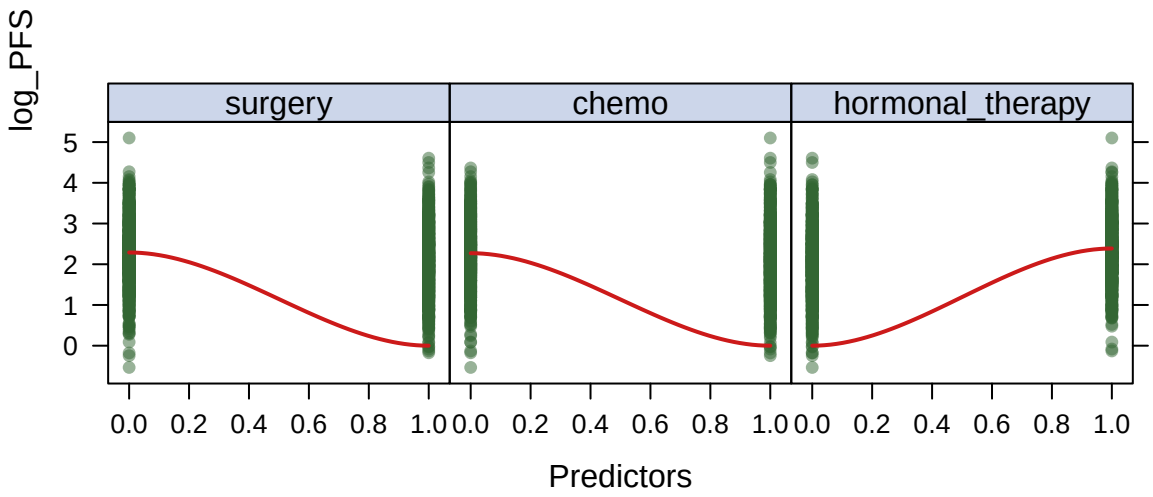
```
load("data/datA.RData")

set.seed(2222)

datA_split = initial_split(datA, prop = 0.8)

#extract the training and test data
training_datA = training(datA_split)
testing_datA = testing(datA_split)

#training data
x_trainA = model.matrix(log_PFS ~ . - id, training_datA)[, -1]
y_trainA = training_datA$log_PFS

#test data
x_testA = model.matrix(log_PFS ~ . - id, testing_datA)[, -1]
y_testA = testing_datA$log_PFS
```

Exploratory Data Analysis

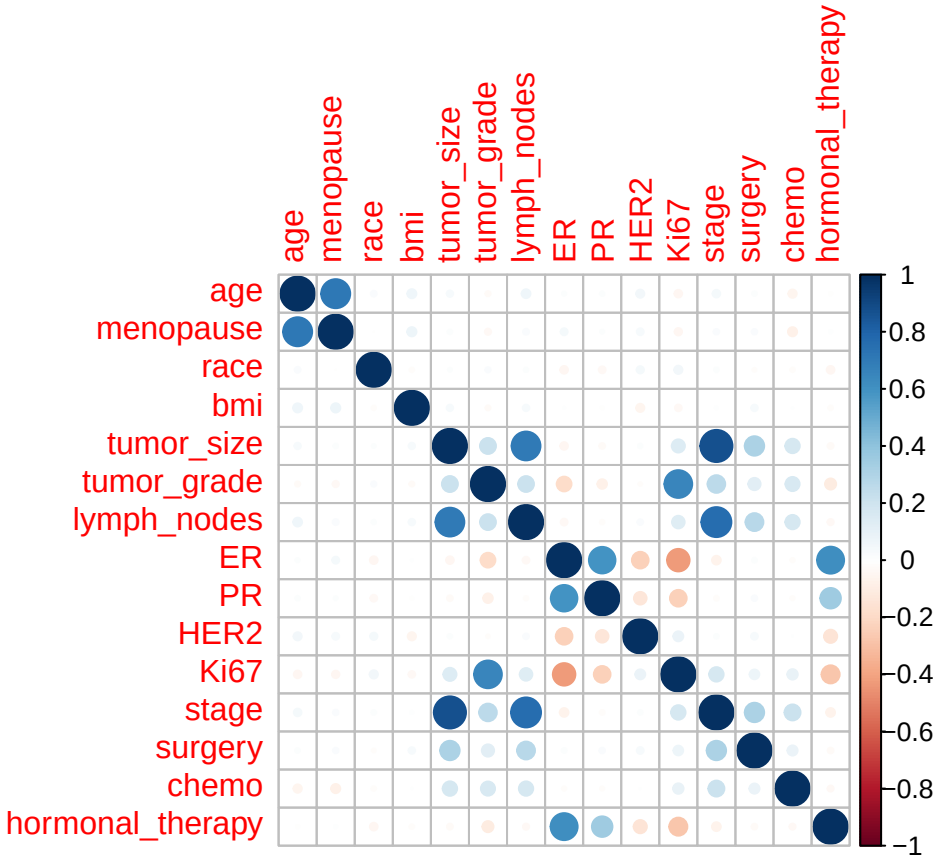
```r
# Changing the plot display
theme1 <- trellis.par.get()
theme1$plot.symbol$col <- rgb(.2, .4, .2, .5)
theme1$plot.symbol$pch <- 16
theme1$plot.line$col <- rgb(.8, .1, .1, 1)
theme1$plot.line$lwd <- 2
theme1$strip.background$col <- rgb(.0, .2, .6, .2)
trellis.par.set(theme1)

# Feature plots to look at trends between predictors and the outcome (log_PFS)
featurePlot(x = training_datA[ ,2:16],
            y = training_datA[ ,17],
            plot = "scatter",
            span = .5,
            labels = c("Predictors","log_PFS"),
            type = c("p", "smooth"),
            layout = c(3, 2))
```




HER2
Ki67
stage
lymph_nodes
ER
PR
log_PFS
Predictors




```
# Calculates the correlation matrix for all the predictors
corrplot(cor(x_trainA), method = "circle", type = "full")
```



Results: Report the model. Interpret the model and assess its performance.

Part II

About a decade later, Researcher B conducted a separate study enrolling 2000 breast cancer patients diagnosed between 2000 and 2005, using the same study design and variable definitions. Like Researcher A's study, all patients were followed until disease progression or death. Researcher B's data are stored in datB.RData. Researcher A would like to evaluate whether the prediction model built on the original data generalizes well to Researcher B's dataset. Does the model generalize well? If yes, provide justification. If not, build a new prediction model for the 2000–2005 cohort and discuss what has changed in prediction.

Import and partition data using 'tidymodels' package

```
load("data/datB.RData")

set.seed(2222)

datB_split = initial_split(datB, prop = 0.8)

#extract the training and test data
training_datB = training(datB_split)
testing_datB = testing(datB_split)
```

```r
#training data
x_trainB = model.matrix(log_PFS ~ . - id, training_datB)[, -1]
y_trainB = training_datB$log_PFS

#test data
x_testB  = model.matrix(log_PFS ~ . - id, testing_datB)[, -1]
y_testB = testing_datB$log_PFS
```

Exploratory Data Analysis

```r
# Feature plots to look at trends between predictors and the outcome (log_PFS)
featurePlot(x = training_datB[ ,2:16],
            y = training_datB[ ,17],
            plot = "scatter",
            span = .5,
            labels = c("Predictors","log_PFS"),
            type = c("p", "smooth"),
            layout = c(3, 2))
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.181e-15

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
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## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

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## reciprocal condition number 0
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## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 0.985

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 0.000225

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.985
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.015

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 0.985

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 0.000225

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.985

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.015

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 0.985

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 0.000225

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.985

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.015
```

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## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
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## 0.985

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## radius 0.000225

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## all data on boundary of neighborhood. make span bigger

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## There are other near singularities as well. 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 2.1403e-15

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

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## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

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## reciprocal condition number 0
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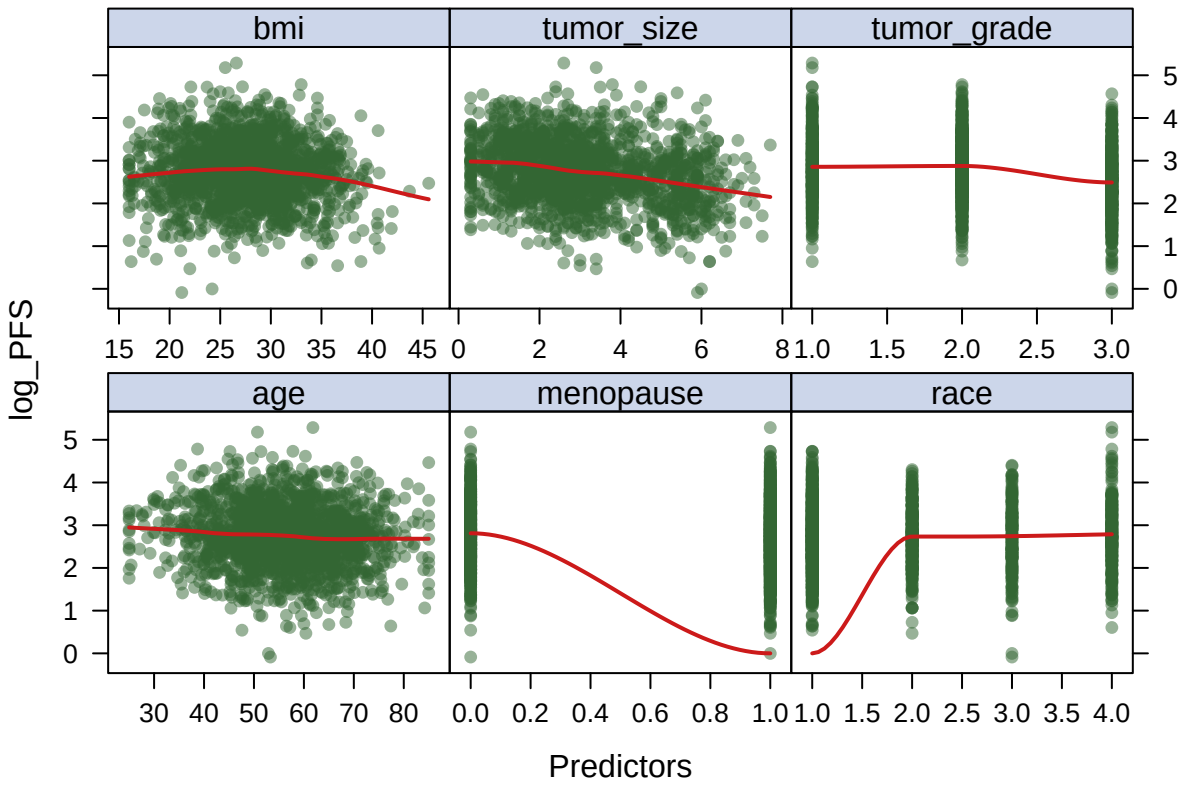


```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number -0
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1

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## reciprocal condition number -0

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## neighborhood radius 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number -0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
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## zero-width neighborhood. make span bigger

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## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.6438e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

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## neighborhood radius 1.005

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## reciprocal condition number 0

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## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005
```
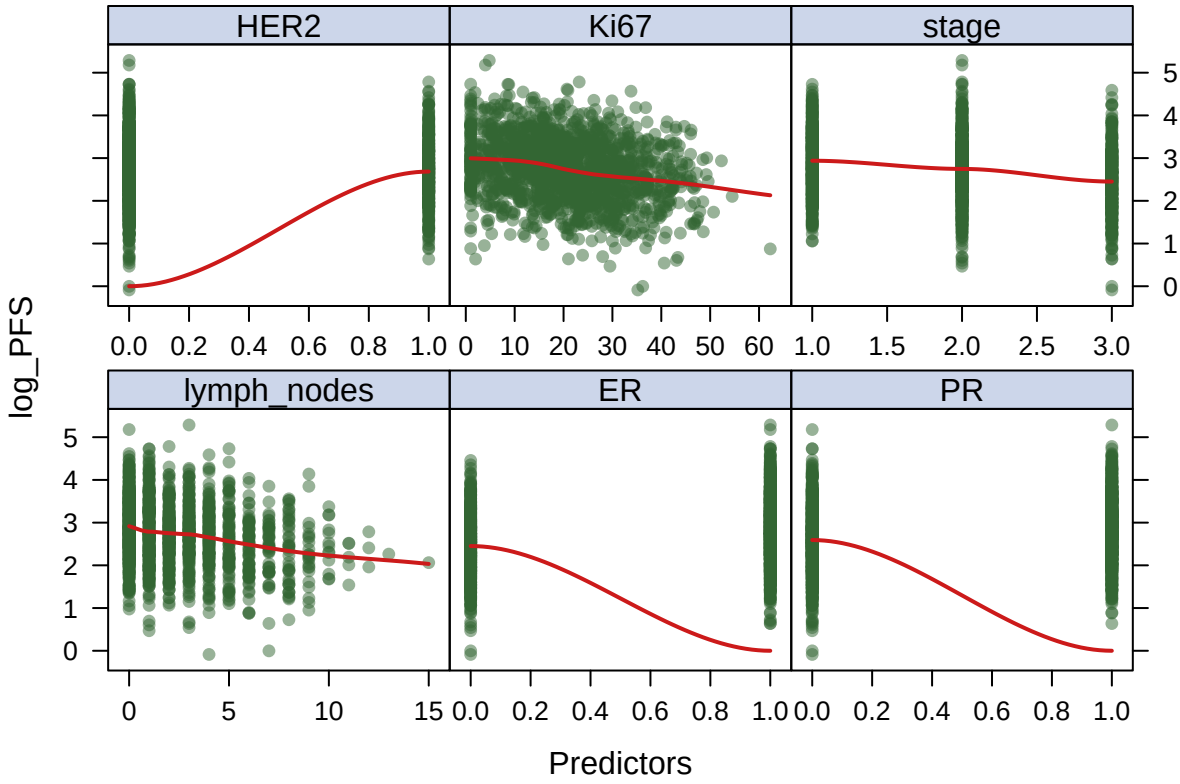

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.6438e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 0.99

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.0201
```




```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 1.01

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005
```

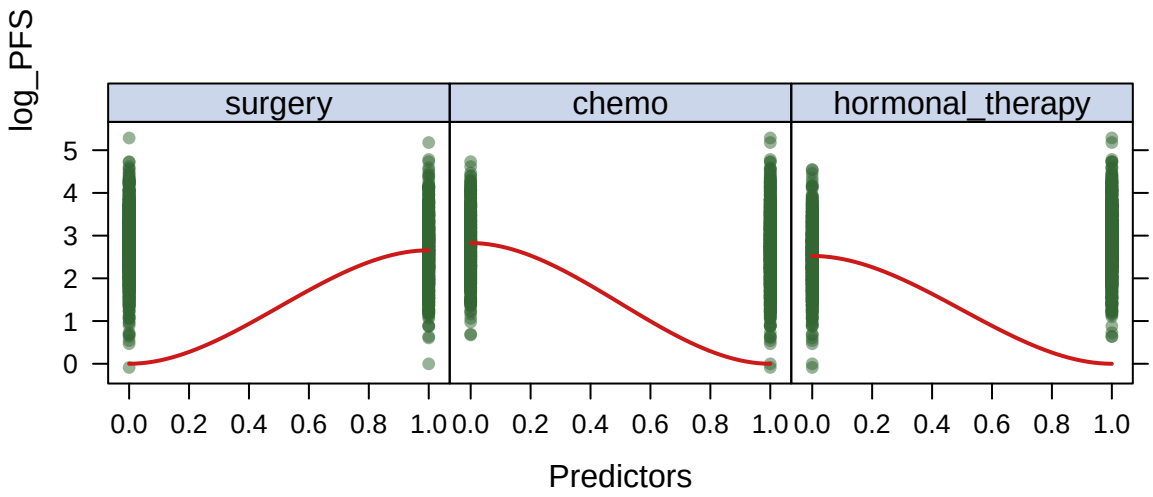
```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.1167e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 5.0227e-16
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.1167e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005
```

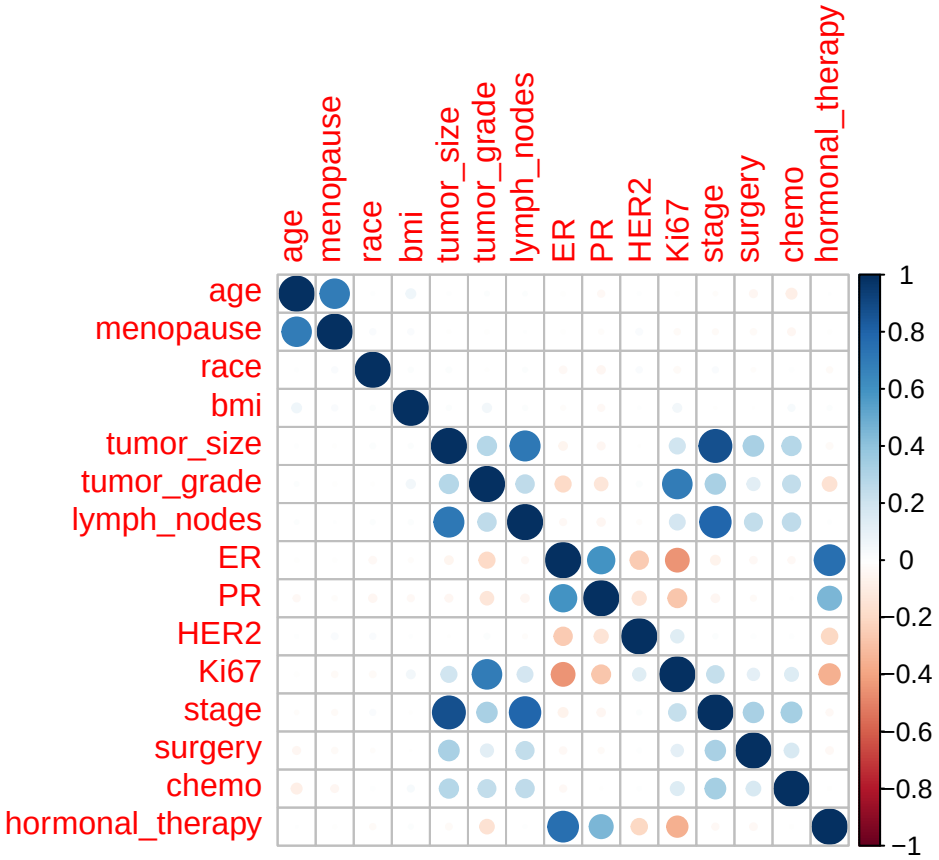
```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.3994e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at -0.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 1.3994e-14

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, : at
## 1.005

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## radius 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## There are other near singularities as well. 2.5e-05

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## zero-width neighborhood. make span bigger
```




```
# Calculates the correlation matrix for all the predictors
corrplot(cor(x_trainB), method = "circle", type = "full")
```


age
menopause
race
bmi
tumor_size
tumor_grade
lymph_nodes
ER
PR
HER2
Ki67
stage
surgery
chemo
hormonal_therapy
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1